

The GNoME Low- κ_L Screen

Architecture, physics, and the crystallographic structure of 33,323 candidates

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Abstract

A three-model CGCNN ensemble predicts bulk and shear modulus for every material in a 554,054-entry snapshot of Google DeepMind’s GNoME discovery database; the paper’s own Slack-model physics turns those moduli into a predicted lattice thermal conductivity κ_L ; and 16,299 candidates clear the screening threshold of $\kappa_L \leq 1 \text{ W m}^{-1} \text{ K}^{-1}$. That screen, and the model behind it, are described in full in `docs/method.tex`. This document is the follow-on: having a candidate list is not the same as understanding it, so every one of those 33,323 scored candidates is here tagged with its crystallographic space group (read directly from GNoME’s own structure metadata) and its constituent elements, then broken down two ways — across the full screen, and across the 11,612 oxide-containing candidates specifically — to ask what the low- κ_L region of chemical space actually looks like. Two findings stand out. First, low- κ_L candidates are not uniformly distributed across chemistry: halogens (Br, I, Cl, F) are markedly *enriched* among low- κ_L hits (up to $1.9\times$ their background rate) while oxygen is markedly *depleted* ($0.27\times$), consistent with soft, heavy-halide lattices having lower sound velocities and Debye temperatures than tightly-bonded oxides. Second, both the general screen and the oxide subset skew heavily toward low-symmetry space groups (triclinic and monoclinic account for two-thirds of all candidates), a pattern this document quantifies precisely rather than leaves as an impression. Finally, we identify the single strongest candidate in every crystal system and every space group, for both populations — ranked not by the raw point-estimate κ_L , but by its 95th-percentile Monte Carlo value, since ranking $> 30\text{k}$ candidates by the point estimate alone turns out to be a textbook winner’s-curse setup: the top point-estimate oxide and the runner-up general candidate both carry $19\text{--}22\times$ p95/p05 uncertainty ratios, indistinguishable from noise rather than a confident low- κ_L prediction. Re-ranked this way, **CrSi₃Cl₁₆** (triclinic, $P1$) holds its #1 spot in the general screen (p95 = $0.0094 \text{ W m}^{-1} \text{ K}^{-1}$, $7.6\times$ ratio); the oxide leaderboard reorders, with **COF₁₄** (monoclinic, $P2_1$) taking the top spot at a much tighter $2.3\times$ ratio.

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1 Background and pipeline

The full pipeline, trained model, and validation are described in `docs/method.tex`; this section recaps only what is needed to read the results below.



Every material in this document already passed the paper’s own three-stage funnel (bandgap, thermodynamic stability, no radioactive elements) before being scored:

Table 1: The screening funnel, this project’s current 554,054-material GNoME snapshot.

Stage	Candidates remaining
GNoME snapshot	554,054
Bandgap [0.1, 3.0] eV, decomposition energy ≤ 0	37,889
+ no radioactive elements	33,323 (scored by CGCNN)
+ $\kappa_L \leq 1 \text{ W m}^{-1} \text{ K}^{-1}$ (own model)	16,299 (“low- κ_L ” below)

2 Model and physics summary

2.1 The network: CGCNN

Each crystal becomes a graph — atoms as nodes, bonds as edges — and the network predicts K_{VRH} and G_{VRH} (Voigt–Reuss–Hill bulk and shear modulus) from it directly, with no DFT calculation involved.

- **Nodes.** Each atom is a 92-dimensional vector of tabulated element properties (group, period, electronegativity, radius, ...), looked up by atomic number, not learned.
- **Edges.** Every atom’s 12 nearest neighbours within 8 Å, respecting periodic boundary conditions, with the bond distance expanded onto 41 Gaussian basis functions rather than used as a single raw number.
- **Convolution (gated message passing), three layers.** For atom i with neighbours j and bond feature $\mathbf{e}_{ij}$:

$$\mathbf{v}_i^{(t+1)} = \mathbf{v}_i^{(t)} + \sum_j \underbrace{\sigma(\mathbf{z}_{ij}^{(t)} \mathbf{W}_f + \mathbf{b}_f)}_{\text{gate}} \odot \underbrace{g(\mathbf{z}_{ij}^{(t)} \mathbf{W}_s + \mathbf{b}_s)}_{\text{message}}, \quad \mathbf{z}_{ij}^{(t)} = \mathbf{v}_i^{(t)} \oplus \mathbf{v}_j^{(t)} \oplus \mathbf{e}_{ij} \quad (1)$$

with σ a sigmoid (how much of a neighbour’s message to admit) and g a softplus (what that message says). Shared weights across every edge are what make this a *convolution*; the residual term lets gradients reach every layer.

- **Pooling.** The crystal-level vector is the *mean* over its atoms (not the sum), so the prediction does not change if the unit cell is doubled — essential for an intensive property like a modulus.
- **Ensembling.** Three independently-initialised models per target, averaged in log space; their spread is a free per-crystal uncertainty estimate with no equivalent in a single pre-trained model.

2.2 The physics: Slack model

Everything from (K, G) to κ_L is closed-form physics, unchanged from the paper, driven by our own predicted moduli:

$$v_{\text{long}} = \sqrt{(K + \frac{4}{3}G)/\rho}, \quad v_{\text{trans}} = \sqrt{G/\rho} \quad (2)$$

$$\gamma = \frac{3(1 + \nu)}{2(2 - 3\nu)}, \quad \nu = \frac{(v_{\text{long}}/v_{\text{trans}})^2 - 2}{2(v_{\text{long}}/v_{\text{trans}})^2 - 2} \quad (3)$$

$$\kappa_L = \frac{G \cdot v_{\text{sound}} \cdot V^{1/3}}{N_{\text{atoms}} \cdot 300} e^{-\gamma} \quad (\text{Eq. 2 of the PINK paper, } \delta = 1) \quad (4)$$

where ν is Poisson’s ratio and γ the Grüneisen parameter — **both derived entirely from the predicted K and G** , the same formula the paper’s own released `pink_predict.py` uses for real screening (its Table 1 validation numbers use a different, literature-looked-up γ instead; see `docs/method.tex`, Section 7.4, for why that distinction matters and is not a bug in this pipeline).

Where the space-group data in this document comes from

Every Space Group / Space Group Number / Crystal System value below is read directly from GNoME’s own `stable_materials_summary.csv` — it is part of the structure metadata GNoME itself computed when the material was generated, joined onto our scored candidates by `MaterialId`. No new structure analysis was run for this document; 0.8–0.9% of candidates have no matching summary row (a small gap in the upstream file, not something this project introduces) and are reported, not silently dropped.

3 Crystallographic structure of the full screen

Table 2: Crystal system distribution, all 33,323 scored candidates (`scripts/cgcnn/17_spacegroups_general.py`). 264 candidates (0.8%) have no matching space-group row.

Crystal system	Total candidates	Low- κ_L candidates	% low- κ_L
Monoclinic	11,026	4,726	42.9%
Triclinic	10,761	6,802	63.2%
Orthorhombic	4,394	1,885	42.9%
Trigonal	4,151	1,632	39.3%
Tetragonal	1,352	575	42.5%
Hexagonal	749	298	39.8%
Cubic	626	254	40.6%

Triclinic candidates clear the threshold at nearly $1.5\times$ the rate of every other crystal system (63.2% vs. 39–43%) — the lowest-symmetry structures are disproportionately likely to be good κ_L candidates, which is physically sensible: low symmetry usually means a more complex, less uniform bonding environment, which scatters phonons more effectively.

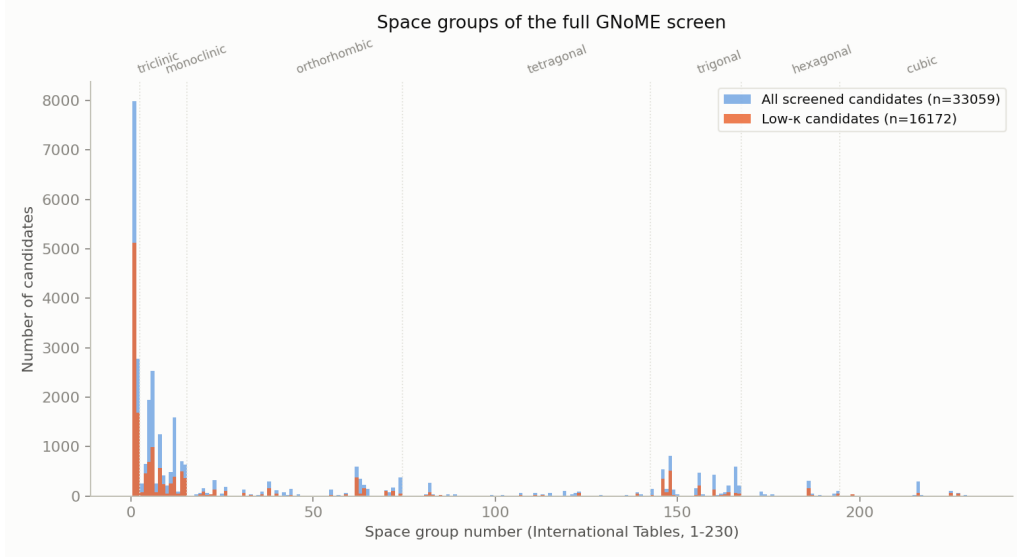


Figure 1: Space-group-number histogram, full screen. All candidates (blue) vs. the 16,299 low- κ_L candidates (orange), with the seven crystal-system windows marked along the top axis.

3.1 Element composition and enrichment

For every element appearing anywhere in the screen, we compute how many candidates contain it and define an *enrichment ratio*: the fraction of low- κ_L candidates containing that element, divided by its fraction of the whole screen. Enrichment > 1 means that element’s compounds skew toward low κ_L more than average; < 1 means the opposite.

Table 3: Most enriched and most depleted elements among low- κ_L candidates (elements with ≥ 30 total candidates, to avoid noisy small counts; `scripts/cgcnn/18_element_frequency.py`).

Element	Most enriched		Element	Most depleted	
	Total	Enrichment		Total	Enrichment
Br	3,319	1.88×	Gd	776	0.17×
I	3,221	1.86×	O	11,612	0.27×
Cl	1,378	1.78×	W	1,357	0.29×
Cs	3,628	1.73×	Mo	1,809	0.35×
Rb	3,863	1.66×	B	858	0.37×
C	711	1.64×	N	1,239	0.40×
F	2,579	1.62×	Ca	1,455	0.42×
K	3,144	1.50×	Sr	2,280	0.50×
Zr	1,240	1.43×	Rh	998	0.56×
Se	3,815	1.40×	Mg	757	0.57×

Halogens up, oxygen down

The four halogens present in appreciable numbers (Br, I, Cl, F) are all among the top seven most-enriched elements, alongside the large alkali cations they typically pair with (Cs, Rb, K). Oxygen, despite being the single most common element in the whole screen (11,612 of 33,323 candidates, 34.8%), is the second most *depleted* among low- κ_L candidates at $0.27\times$. This is physically consistent: heavy, soft, highly polarizable halide lattices have lower sound velocities and Debye temperatures than the stiff, tightly-bonded lattices oxygen tends to form, and lower Debye temperature drives κ_L down directly through Eq. 3.

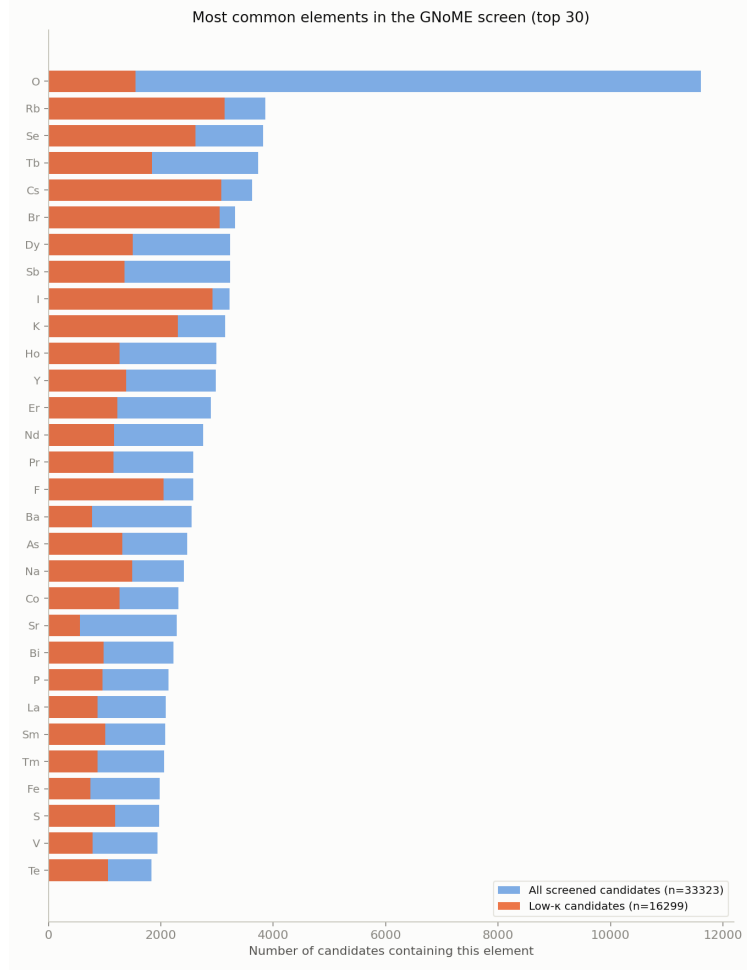


Figure 2: The 30 most common elements in the screen, all candidates (blue) vs. low- κ_L candidates (orange).

4 The oxide subset

11,612 of the 33,323 candidates (34.8%) contain oxygen (`scripts/cgcnn/15_filter_oxides.py`), identified via pymatgen’s composition parser rather than a substring search (which would wrongly flag Os and Og). Only 1,545 of those 11,612 (13.3%) clear the low- κ_L threshold — versus 48.9% for the screen as a whole — consistent with Section 3.1’s finding that oxygen is the second most depleted element.

Table 4: Crystal system distribution, oxide candidates only (`scripts/cgcnn/16_spacegroups_oxides.py`). 102 candidates (0.9%) have no matching space-group row.

Crystal system	Total oxide candidates	Low- κ_L candidates	% low- κ_L
Monoclinic	4,280	561	13.1%
Triclinic	3,373	500	14.8%
Trigonal	1,747	155	8.9%
Orthorhombic	1,415	184	13.0%
Tetragonal	419	94	22.4%
Hexagonal	176	18	10.2%
Cubic	100	22	22.0%

Table 5: Stoichiometry pattern (reduced cation:oxygen ratio; `15_filter_oxides.py`’s own classification, a formula-ratio tag, not a DFT-verified structure). No exact ABO₃ (perovskite) or simple binary oxide survived the funnel — GNoME’s discovered materials skew toward complex, previously-uncatalogued compositions.

Pattern	Total candidates	Low- κ_L candidates
Complex oxide (4 elements)	6,925	1,114
Complex oxide (5 elements)	4,110	230
Ternary oxide (A–B–O, other ratio)	548	185
A ₂ B ₂ O ₇ -type (pyrochlore-like)	21	2
AB ₂ O ₄ -type (spinel-like)	8	3

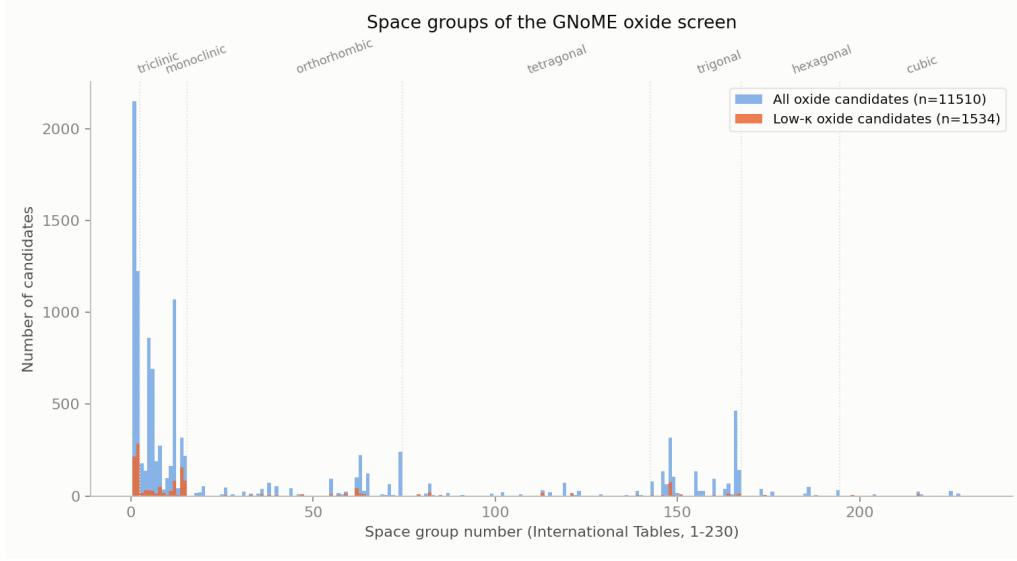


Figure 3: Space-group-number histogram, oxide candidates only. Same overlay/axis convention as Figure 1.

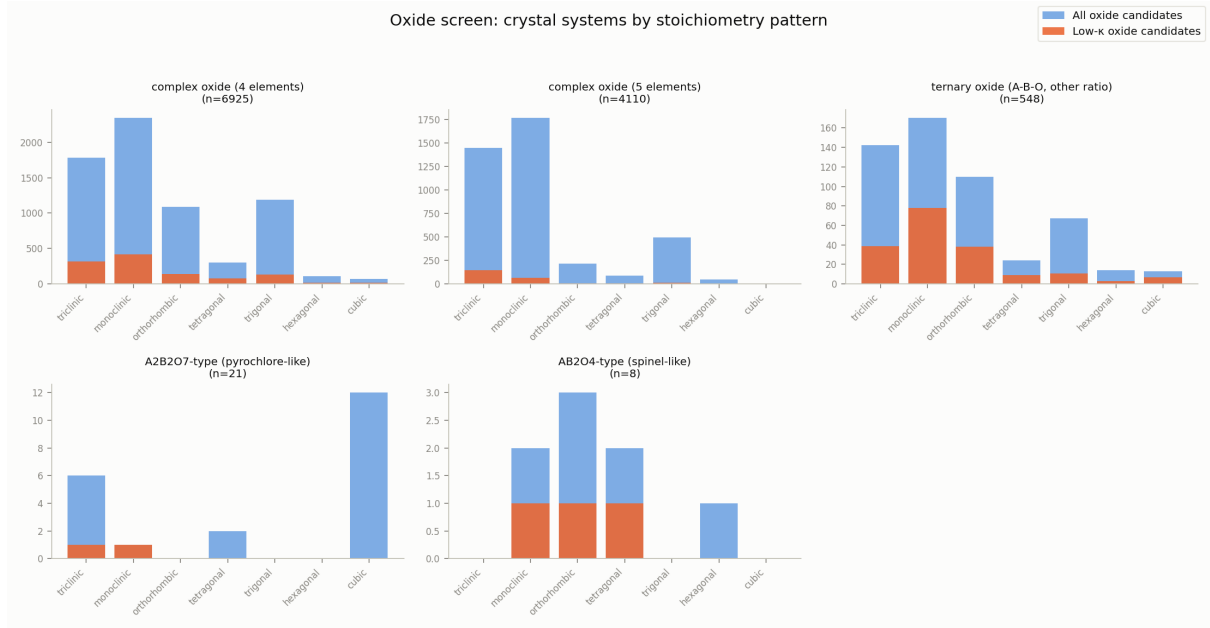


Figure 4: Crystal-system distribution within each stoichiometry pattern. Each panel has its own y -scale so the 8-row AB_2O_4 pattern is not crushed flat next to the $> 4,000$ -row complex-oxide patterns. The 21 pyrochlore-like candidates are notably cubic-dominated (12/21) — the canonical pyrochlore space group is cubic $Fd\bar{3}m$ — but none of those 12 cubic pyrochlores clear the low- κ_L threshold.

5 The strongest candidates

“Best” below means lowest κ_L $p95$ — the 95th-percentile Monte Carlo value from Section 2.2 — not the raw point estimate. Every row is shown regardless of whether it actually clears the $\leq 1 \text{ W m}^{-1} \text{ K}^{-1}$ threshold, so a weak group is shown honestly rather than omitted.

Why rank by p95, not the point estimate

Ranking 33,323 candidates by lowest point-estimate κ_L and taking the minimum is a text-book winner’s-curse setup: a candidate needs either a genuinely low κ_L or just a wide, noisy ensemble disagreement to land at the top, and the two look identical in a point estimate alone. Checked directly on this data: ranking by the point estimate instead of p95, the top oxide (ReAsOF₁₀) and the second-best general candidate (HgWCl₁₀) both carry p95/p05 ratios of 19–22× — solidly in the “worth a DFT check” range by this document’s own convention (Section 3), not the confident end of it. Ranking by p95 instead asks a more conservative question — *is this candidate still predicted low- κ_L even in the model’s own pessimistic scenario* — and rewards confident, tightly-bounded predictions over lucky, noisy ones. The ratio column below makes that confidence visible rather than hiding it behind a single number.

Table 6: Best candidate per crystal system by κ_L p95, oxide vs. general screen. Ratio = p95/p05, this candidate’s own Monte Carlo spread — lower is more confident.

Crys. sys.	Formula	Oxide subset		Formula	Full screen	
		Sp. grp.	p95 (ratio)		Sp. grp.	p95 (ratio)
Triclinic	CuSCl6O	$P\bar{1}$	0.0756 (4.3×)	CrSi3Cl16	$P1$	0.0094 (7.6×)
Monoclinic	COF14	$P2_1$	0.0663 (2.3 ×)	TaSb(As2Cl13)2	Cm	0.0191 (2.0 ×)
Orthorhombic	ReOF5	$Pnma$	0.0636 (6.2×)	SbP2Cl13	$Cmc2_1$	0.0125 (2.4×)
Tetragonal	AgN5O14	$I\bar{4}$	0.1119 (7.5×)	POsCl10	$P4/n$	0.0163 (4.2×)
Trigonal	S24ClO3	$R3m$	0.0684 (3.8×)	FeSeBr18	$R\bar{3}$	0.0264 (2.2×)
Hexagonal	Rb6BaO4	$P6_3mc$	0.4867 (3.7×)	Rb3Co2Br9	$P6_3/mmc$	0.1046 (4.8×)
Cubic	Fe3Br4O	$P2_13$	0.5684 (11.5×)	CsRb2BiTe3	$P2_13$	0.1079 (2.6×)

Bold marks each population’s single best candidate overall by this conservative ranking. **CrSi₃Cl₁₆ holds its #1 spot even after re-ranking** — its 7.6× ratio, while not the tightest in the table, is still low enough that it stays ahead once every candidate’s own uncertainty is priced in, unlike HgWCl₁₀ (previously #2 by point estimate) which drops out of the top 10 entirely once ranked by p95. Among oxides, the point-estimate winner ReAsOF₁₀ is displaced by COF₁₄ — a much less dramatic κ_L but a 2.3× ratio, the tightest of any oxide crystal-system winner.

Table 7: Top 10 overall (one row per space group, ranked by κ_L p95), oxide subset. Full 134-row table: `gnome_oxide_best_by_spacegroup.csv` (also carries the point estimate and p05 for every row).

#	Formula	Space group	Crystal system	p95	Ratio
1	ReOF5	$Pnma$	orthorhombic	0.0636	6.2×
2	COF14	$P2_1$	monoclinic	0.0663	2.3 ×
3	S24ClO3	$R3m$	trigonal	0.0684	3.8×
4	ReAsOF10	$P2$	monoclinic	0.0701	19.3×
5	CeSCl6O	$P2_1/c$	monoclinic	0.0745	2.1 ×
6	CuSCl6O	$P\bar{1}$	triclinic	0.0756	4.3×
7	Rb2Re2OF10	$P2/c$	monoclinic	0.0837	3.9×
8	TlRe2Ag(OF2)4	$Cmm2$	orthorhombic	0.0974	4.6×
9	Na(S8O)3	$P1$	triclinic	0.1103	7.7×
10	AgN5O14	$I\bar{4}$	tetragonal	0.1119	7.5×

Row 4 is the point-estimate winner from before ($\kappa_L = 0.0158$) — still a strong candidate, just

no longer #1 once its own $19.3\times$ uncertainty is weighed against tighter alternatives like row 2 or row 5.

Table 8: Top 10 overall (one row per space group, ranked by κ_L p95), full screen. Full 167-row table: `gnome_screen_best_by_spacegroup.csv`.

#	Formula	Space group	Crystal system	p95	Ratio
1	CrSi3Cl16	$P1$	triclinic	0.0094	$7.6\times$
2	SbP2Cl13	$Cmc2_1$	orthorhombic	0.0125	$2.4\times$
3	BI5	$Pbcm$	orthorhombic	0.0140	$3.0\times$
4	BPI6	$Pnma$	orthorhombic	0.0161	$4.1\times$
5	POsCl10	$P4/n$	tetragonal	0.0163	$4.2\times$
6	TaSb(As2Cl13)2	Cm	monoclinic	0.0191	$2.0\times$
7	YAg3Br28	Pm	monoclinic	0.0193	$2.7\times$
8	Au3IBr12	$P2_1$	monoclinic	0.0210	$4.5\times$
9	P(WCl6)2	$P2_1/m$	monoclinic	0.0212	$4.8\times$
10	AgBr5	$P2_1/c$	monoclinic	0.0218	$2.9\times$

Every survivor here is again a heavy halide (Cl, Br), reinforcing Section 3.1’s enrichment finding independently of which ranking is used — the chemistry signal is robust to the point-estimate/p95 choice even though the individual winners are not.

A Appendix: CSV column reference

Every CSV referenced above shares the columns below. Columns already present in `gnome_screen_all.csv` (from `13_screen_gnome.py`) ride along unchanged; `space_group`, `space_group_number` and `crystal_system` are added by this document’s own scripts (`16_spacegroups_oxides.py`, `17_spacegroups_general.py`).

Identity

Column	Meaning
<code>material_id</code>	GNoME’s own identifier for the material.
<code>formula</code>	Reduced chemical formula.
<code>stoichiometry_pattern</code>	(oxide files only) reduced cation:oxygen ratio family — see Section 4’s table.
<code>is_low_kappa_candidate</code>	True if $\kappa_L \leq 1 \text{ W m}^{-1} \text{ K}^{-1}$.
<code>space_group</code>	/ Read from GNoME’s own structure metadata — see the keybox in Section 2.2.
<code>space_group_number</code>	
<code>/ crystal_system</code>	

Structural quantities (read from the primitive cell, not predicted)

Number of Atoms	Atom count in the primitive cell.
Volume (Å ³)	Primitive-cell volume, Å ³ .
Density (g cm ⁻³)	Mass ÷ volume.
Atomic mass (amu)	Total composition weight of the primitive cell.

Model predictions (the 3-model CGCNN ensemble mean, linear GPa)

<code>K_VRH_pred</code>	/ Predicted bulk / shear modulus (GPa).
<code>G_VRH_pred</code>	
<code>K_VRH_spread_log10</code>	/ Ensemble disagreement, in log ₁₀ (GPa) — a per-crystal uncertainty estimate.
<code>G_VRH_spread_log10</code>	

Derived physics (Slack model, Section 2.2) and Monte Carlo uncertainty

Poisson ratio	/ ν, γ above.
Gruneisen parameter	
<code>Kappa_cal</code> (W m ⁻¹ K ⁻¹)	The predicted κ_L — the number the whole screen thresholds on.
<code>Kappa_cal_p05</code> / <code>_p50</code> / <code>_p95</code>	5th/50th/95th percentile of κ_L across 2,000 Monte Carlo draws of K and G (width set by the ensemble spread above). A tight interval means the ensemble agreed; a wide one means the candidate is worth a DFT check before trusting.